

Name: _____

Registration: _____

(10 pts.) Draw the block diagram of the ROM_{16x8}. Label all the inputs and outputs appropriately.

```
module ROM16x8 (cs, addrb, datab, read_en);

    input [3:0] addrb;
    output [7:0] datab;
    input cs, read_en;

    wire [7:0] datab1, datab2;
    wire [1:0] OUT;

    assign OUT[0] = cs && ~addrb[3];
    assign OUT[1] = cs && addrb[3];

    ROM1_8x8 rc1 (OUT[0], addrb[2:0], datab1, read_en);
    ROM2_8x8 rc2 (OUT[1], addrb[2:0], datab2, read_en);

    assign datab = (cs && read_en)?datab1 | datab2:8'bz;

endmodule
```

```
module ROM1_8x8 (cs, addrb, datab, read_en);

    input [2:0] addrb;
    output [7:0] datab;
    input read_en, cs;

    reg [7:0] datab;

    always @ (*)
        if (read_en & cs)
            case (addrb)
                0: datab = 21;
                1: datab = 255;
                2: datab = 33;
                3: datab = 99;
                4: datab = 127;
                5: datab = 13;
                6: datab = 10;
                7: datab = 88;
            endcase
        else
            datab = 8'b0;

endmodule
```

```
module ROM2_8x8 (cs, addrb, datab, read_en);

    input [2:0] addrb;
    output [7:0] datab;
    input read_en, cs;

    reg [7:0] datab;

    always @ (*)
        if (read_en & cs)
            case (addrb)
                0: datab = 28;
                1: datab = 38;
                2: datab = 48;
                3: datab = 58;
                4: datab = 68;
                5: datab = 78;
                6: datab = 88;
                7: datab = 98;
            endcase
        else
            datab = 8'b0;

endmodule
```

(10 pts.) Simulate the ROM_16x8 using the below testbench. And write the output in the format specified by \$monitor.

```
module tst_ROM_16x8;

    reg [3:0] addrb;
    wire [7:0] datab;
    reg cs, read_en;

    ROM_16x8 rom (cs, addrb, datab, read_en);

    initial begin
        #5 addrb = 0;           //0 000
        #10 addrb = 7;          //0 111
        #10 addrb = 8;          //1 000
        #5 addrb = 15;          //1 111
        #5 addrb = 14;          //1 110
        #5 addrb = 11;          //1 011
        #5 addrb = 12;          //1 100
        #5 addrb = 9;           //1 001
    end

    initial begin
        cs = 1;
        read_en = 1;
    end

    initial
        $monitor ("%d: data: %d", $time, datab);

endmodule
```